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Hiring Committee
Flatiron Institute - Simons Foundation,
New York, NY

Dear Members of the Selection Committee,

I am writing to apply for the Summer 2026 Internship Program at the Center for Computational Biology (CCB) at the Flatiron Institute. I am currently a master's student in Mathematical Sciences (Statistics concentration) at Georgia Southern University, with research experience spanning statistical modeling, stochastic processes, and computational analysis of biological systems. I am highly motivated to contribute to CCB's mission of developing rigorous theoretical and computational tools to understand complex biological processes across scales.

What excites me most about CCB is its deep integration of mathematics, statistics, computation, and biological insight. The Center's emphasis on mechanistic modeling, inference, and theory-driven data analysis strongly aligns with my training and research interests. I am particularly interested in projects within **Statistical Biophysics**, **Genomics**, **Developmental Dynamics**, and **Biophysical Modeling**, where probabilistic modeling, simulations, and data-driven inference are central to advancing biological understanding.

In my current research role at the Clonal Redesign Lab at Moffitt Cancer Center, I develop and analyze stochastic models of chromosomal mis-segregation and karyotype evolution in cancer cell populations. My work involves simulating thousands of cell-division trajectories, estimating transition probabilities, and comparing simulated outcomes with experimental data using distance-based statistics and hypothesis testing. This experience has given me a strong foundation in stochastic processes, non-equilibrium systems, simulation-based inference, and translating abstract mathematical models into biologically interpretable results.

Previously, I supported statistical modeling and data analysis for clinical research projects at a clinical trials center, where I worked on experimental design, randomization strategies, and data validation pipelines. These experiences strengthened my appreciation for careful data curation, reproducibility, and collaboration across disciplinary boundaries, values that are clearly central to CCB's research culture.

Technically, I am proficient in R and Python for statistical analysis, simulation, and visualization, with experience in regression modeling, Bayesian methods, survival analysis, and computational workflows for large datasets. I am particularly interested in expanding my skills in high-performance computing, advanced numerical methods, and theory-driven inference under the mentorship of CCB scientists. Beyond technical skills, I bring strong curiosity, attention to detail, and enthusiasm for collaborative, open-ended research.

I am excited by the opportunity to contribute meaningfully to ongoing CCB research projects while learning from a vibrant community of scientists at the Flatiron Institute. I would be honored to be considered for the Summer 2026 Internship Program and welcome the opportunity to further discuss how my background and interests align with CCB's research goals.

Thank you very much for your time and consideration.

Sincerely,
Daniel Olofin